Brachytherapy 14 (2015) 726–733

BRACHYTHERAPY

Prescribing to tumor apex in episcleral plaque iodine-125 brachytherapy for medium-sized choroidal melanoma: A single-institutional retrospective review

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ABSTRACT

PURPOSE: To report an institutional experience with episcleral plaque brachytherapy for medium-sized uveal melanoma. Variations in prescription dose point and dose rate were compared with Collaborative Ocular Melanoma Study (COMS) Group.

METHODS AND MATERIALS: A retrospective review was performed for 116 patients treated with iodine-125 plaque brachytherapy. About 85 Gy was prescribed to either the tumor apex (108 patients) or at 5 mm (8 patients) with dose rate ranging from 50.6 to 98.2 cGy/h. Patients were followed up for local tumor control, eye preservation, and vision retention. Dose and dose rate to tumor and sensitive structures were calculated. Multivariate and univariate analyses were performed to investigate correlation between clinical outcomes and dose/dose rate variables.

RESULTS: Patients in this study were slightly older with worse visual acuity at baseline, but tumor size and position and ratio of ciliary body involvement were comparable to COMS population. Outcomes data were comparable to COMS: 95.3% local tumor control at 5 years and 77.7% vision preservation at 3 years. Only 4 patients needed enucleation because of tumor growth. Significant correlation was found between enucleation and tumor height and maximal scleral dose/dose rate as well as vision retention and tumor height and macula dose/dose rate.

CONCLUSIONS: For tumors with <5 mm height, prescribing to tumor apex enabled to decrease dose to all sensitive structures without any loss of local control. Although dose rate was lowered to 50.6 cGy/h from the American Brachytherapy Society guidelines (60–105 cGy/h) because of limited availability of operating room (i.e., weekly), there was no difference in either local tumor control or complications.

Keywords: Uveal melanoma; Episcleral plaque; ^{125}I; Dose rate

Introduction

Uveal melanoma, although rare, is the most common primary intraocular malignancy in adults and represents a threat to both vision and life of the patient. Treatment options for uveal melanoma include enucleation, photocoagulation, eyewall resection, external beam radiotherapy (protons), and plaque brachytherapy (1–5). Historically, large tumors (>10 mm in height or >16 mm in the largest basal diameter) have been treated with enucleation. Gamma Knife radiosurgery has been also considered as an alternative treatment option to enucleation for large uveal melanoma (6). Small tumors (≤2.5 mm in height and ≤10 mm in the largest dimension) may be followed closely with most patients demonstrating no tumor growth (7).

For progressive small tumors and medium-sized tumors, some form of treatment is recommended. Enucleation provides excellent control at the cost of poor cosmesis and

Received 7 November 2014; received in revised form 21 April 2015; accepted 4 May 2015.

Conflict of interest: None to report.

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1538-4721/$ - see front matter
http://dx.doi.org/10.1016/j.brachy.2015.05.002

immediate loss of vision. Plaque brachytherapy was established by the landmark clinical trial performed by the Collaborative Ocular Melanoma Study (COMS) Group as an equivalent alternative to enucleation. COMS demonstrated equivalence between immediate enucleation and plaque brachytherapy in terms of survival for patients with medium-sized uveal melanomas (8). The potential to preserve vision and avoid cosmetically disfiguring surgery has led to the widespread acceptance of this approach for suitable patients.

However, loss of vision and enucleation (either for tumor progression or ocular complications) remain significant problems after iodine-125 (^{125}I) plaque brachytherapy (9). More broadly, enucleation can be performed for adverse effects of radiation therapy, which include ocular pain, among other problems. Radiation dose and dose rate are important factors predicting tumor control (10) and likely are also predictive of radiation retinopathy and ocular pain after brachytherapy. This study reports our institutional experience with uveal melanoma using the ^{125}I episcleral brachytherapy technique with particular attention to the variation in the prescribed dose points, duration of implant, and dose rate.

Methods and materials

A retrospective chart review was performed for all patients treated with ^{125}I plaque brachytherapy for uveal melanoma at the University of Arizona from 1990 to 2010. A total of 176 patients were identified. Records on 22 patients treated before 1998 lacked detailed dosimetric records and were excluded from this study. These 154 patients were treated by two ophthalmology groups, and most patients (140) were treated by a single retinal specialist/ocular oncologist. Hence, patients treated by the second ophthalmology group were excluded from analysis to eliminate variations in treatment and followup patterns depending on each ophthalmologist. Some dosimetric and followup data were missing in 20 of 140 patients. In addition, tumor height was <2.5 mm for 4 patients and considered as small-sized tumor. Finally, the records of the 116 patients with medium-sized tumors were reviewed and compared with the COMS study. This retrospective review study was approved by our institutional review board.

Treatment procedure

Patients diagnosed with medium-sized uveal melanoma were routinely offered either enucleation or ^{125}I plaque brachytherapy. Patients electing brachytherapy were referred to a radiation oncologist. Based on the report from the ophthalmologist office regarding the measurement data of tumor size and distances to critical structures, a medical physicist calculated doses to the prescribed point as well as organs at risks and accordingly constructed a gold plaque using ^{125}I seeds embedded in a silastic insert. The plaque covered the largest basal dimension of tumor with a radial margin of 2−3 mm. The plaque size was determined for each individual patient as one of following six available plaques: 10, 12, 14, 16, 18, and 20 mm. The number of seeds corresponding to each plaque was 5, 8, 13, 13, 21, and 24, respectively. When a notched plaque was used (because of location of the tumor close to the optic nerve), one seed was removed to facilitate the notch, and the location of the remaining seeds was the same as those of a standard plaque. Eight of the 116 patients had tumors located close to the optic nerve and required a notched plaque to provide less than 2-mm margin on the nerve side of the tumor.

For tumors greater than 5 mm in height (41 cases), the dose point for the prescription was always the apex of tumor. For tumors less than 5 mm in height (75 cases), the dose point was either the apex of tumor for 67 cases or 5 mm above the sclera surface for 8 cases. Initially, the prescription point was 5 mm above the sclera surface (following COMS recommendation), but it was later changed to the apex of tumor. In years 1998 and 1999, 7 of 10 patients were prescribed to the 5 mm above the sclera surface and 1 patient in year 2001. Since 2002, all patients have been prescribed to the apex of the tumor if tumor is less than 5 mm in height. Therefore, most treatments had the prescription to the apex of tumor (67 of 75 cases). When the American Association of Physicists in Medicine Task Group-43 (AAPM TG-43) formalism (11) became available for the dose calculation in brachytherapy, 85 Gy was used for the prescribed dose. For earlier patients before AAPM TG-43 formalism was available, 100 Gy was prescribed to the apex of tumor using parameters and formalism provided by the vendor after validation. For those patients, the doses to all points were recomputed based on the AAPM TG-43 formalism with correct parameters. Dose calculation was performed in the brachytherapy module of a commercial Treatment Planning System (TPS) (Pinnacle[3] v9.0; Phillips Medical Systems, Madison, WI). A point source approximation in the updated AAPM TG-43U1 (12) and its supplement (TG-43U1S1 (13)) was used without anisotropic factor being taken into account. Following COMS protocol, the dosimetric influence of the gold plaque or silastic insert was ignored in the dose calculation. Also, the modification of plaque rim was not taken into account for dose calculation in the notched plaque cases. Doses were calculated and reported for the following points: apex of tumor, 5 mm above the sclera, sclera at plaque center (representing the maximal scleral and retinal doses), center of globe, retina on the opposite aspect of the globe (representing the minimal retinal dose), macula, optic disc, and lens. The coordinates of these points (Fig. 1) were determined based on the measurement data from the ophthalmologist by following COMS study (14). To verify computed dose to prescription point and critical structures in the TPS, an inhouse software was developed in MATLAB (release 2012b; The MathWorks, Inc, Natick, MA). This second check software calculates

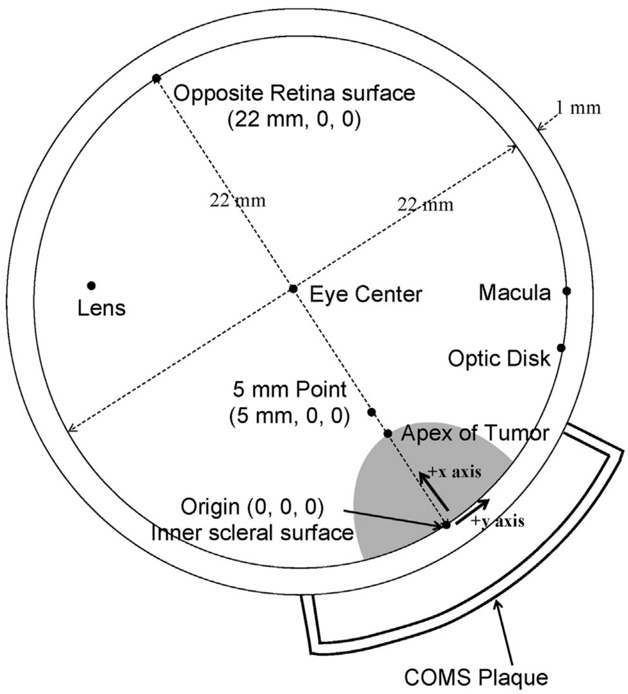

Fig. 1. Collaborative Ocular Melanoma Study eye as a sphere of 12-mm outer radius and location of dose points of interest in a right-handed Cartesian coordinate system. The inner sclera surface is a sphere with 11-mm radius.

coordinates of all source positions and dose points of interest based on plaque size and measured data from the ophthalmologist and then makes dose calculation using the same TG-43U1 parameters as in the TPS for given source strength and implantation time.

The duration of implantation varied ranging from 94.1 h (3.9 days) to 168.0 h (7 days) with a median of 96.0 h, depending on operating room scheduling and the physician's preference. Hence, the range of dose rate at the prescription point was 50.6 to 98.2 cGy/h. The combination between variables of tumor size, tumor location, and plaque duration resulted in considerable differences in dose and dose rate to sensitive structures within the eye.

Patient followup

Patients were followed at increasing intervals (at a minimum of every year) by the operating ocular oncologist for a planned 10 years. At each followup, visual acuity was analyzed and tumor size was measured with ultrasound. All patients were evaluated for local tumor control, distant metastasis, enucleation, and preservation of vision. In this study, local tumor control was considered tumor control. To monitor for the presence of metastatic disease, chest X-ray and liver function tests were performed annually. Preservation of functional vision was defined as visual acuity of better than 20/200. Patients without functional vision at the time of plaque placement were not included in the analysis of vision preservation. Patients with enucleation because of any cause were considered as having no functional vision irrespective of vision preservation before enucleation. Tumor size, tumor location, and plaque duration as well as the dose to critical structures were examined to assess their impact on preservation of vision and enucleation for reasons other than tumor progression.

Statistical analysis

Frequency tables and univariate statistics were computed and reported for the age of patients and other clinical variables. Logistic regression was selected to perform univariate and multivariate analyses on retention of vision, enucleation, and tumor control. An alpha value of 0.05 was established to consider statistical significance. Kaplan−Meier analysis was used to determine survival and estimate the event times of loss of vision, enucleation, and local failure. All analysis was performed using Stata (Windows version 10.1; StataCorp, College Station, TX).

Results

The data of this single institution were compared with those of COMS. The effect of variation in this study for the prescribed dose points, duration of implantation, and dose rate on the dose to the tumor and critical structures was evaluated using patient outcomes data, such as local control, eye preservation, and visual acuity. These clinical outcomes data were also compared with the data of COMS.

Patient age, ciliary body involvement, and baseline visual acuity

Patient age, ciliary body involvement, and baseline visual acuity at the time of plaque insertion are summarized in Table 1. The average age at plaque insertion was 65 years with standard deviation of 11.7 years. Compared with the COMS study, the percentage of patients younger than 50 years (younger patient group) was lower in this study (12.1% vs. 25.7%), whereas higher (38.8% vs. 25.3%) for patients older than 70 years (older patient group). This study has slightly less ciliary body involvement (6.0% vs. 8.5%) compared with COMS study. However, if unknown data in this study were disregarded for analysis, the percentage of lesions without ciliary body involvement was the same (90.7%) between COMS and this study. Median visual acuity at plaque insertion was 20/40; the percentage of good visual acuity (<20/40) patients was slightly lower in this study (58.6% vs. 70.3%). Nonfunctional vision (20/200 or worse) was present in 11 patients (9.5%) at the time of plaque insertion, similar to that of COMS (9.8%). In summary, compared with those in COMS study, the patients in this study were older (median of 65 vs. 61) with slightly worse baseline visual acuity (median of 20/40 vs. 20/32) at the time of plaque brachytherapy. The ciliary body involvement was similar.

Table 1
Comparison of age, ciliary body involvement and baseline visual acuity at the time of plaque brachytherapy between COMS and this study

Characteristics	COMS (%)	Present study number of patients (%)
Age at plaque placement		
<50 y	25.7	14 (12.1)
50–69 y	51.0	57 (49.1)
≥ 70 y	25.3	45 (38.8)
Median	61 years	66 years
Ciliary body involvement		
No	90.7	97 (83.6)
Yes	8.5	7 (6.0)
Unknown	0.0	9 (7.8)
Unavailable	0.8	3 (2.6)
Baseline visual acuity		
≥20/20	33.4	16 (13.8)
<20/20 to 20/40	36.9	52 (44.8)
<20/40 to 20/160	18.9	37 (31.9)
<20/160	9.8	11 (9.5)
Median	20/32	20/40

COMS = Collaborative Ocular Melanoma Study.

Tumor size and height and doses to critical structures

According to the COMS classification for tumor size, all tumors were of medium size (apical height was from 2.5 to 10 mm, and maximum basal dimension was ≤16 mm). The measurement data of tumor size and height as well as the distances to critical structures in this study were compared with those in published COMS study (Table 2). All measured data in this study were similar to those in the COMS study. The difference of median values in the measurements was very small: 0.4 mm for the largest basal dimension; 0.2 mm for apical height of the tumor; and 1.5 mm for macula distance to tumor; furthermore, the median optical disc distance to tumor was the same in both studies. The average ± standard deviation of percentage value difference between two studies was 3.9% ± 3.3%. The maximum percentage value difference was 11.5% (36.8% vs. 48.3%) for the largest basal dimension measurement of the 11–14-mm lesions (Table 2). The dose and dose rate values to the target and critical structures are summarized in Table 3. In particular, the dose to critical structures was compared with those of COMS study. The tumor apex received the same as the prescribed dose of 85 Gy, whereas the dose to 5 mm was reduced to 76.7 Gy, on average. Consistently, the doses to critical structures were also lower in this study compared with those of COMS study in terms of median values. For instance, the macula dose was decreased dramatically from 79 to 50.4 Gy (36.2% reduction). Table 4 shows average dose values for 67 cases in which tumor height was <5 mm and tumor apex was used for prescription point. By changing prescription point from 5 mm above the sclera surface to apex of tumor, the dose at 5-mm point was reduced by 23.0 Gy (26.9%) on average. The dose to critical structures would be decreased accordingly.

Table 2
Comparison of measurements for tumor dimension and height as well as distances to critical structures between COMS and this study

Tumor characteristics	COMS (%)	Present study number of patients (%)
Largest basal dimension		
4.5–8.0 mm	13.5	13 (11.2)
8.1–11.0 mm	34.3	39 (33.6)
11.1–14.0 mm	36.8	56 (48.3)
14.1–16.0 mm	15.4	8 (6.9)
Median	11.5 mm	11.9 mm
Apical tumor height		
2.5–5.0 mm	64.7	75 (64.6)
5.1–7.5 mm	25.2	35 (30.2)
7.6–10.0 mm	10.1	6 (5.2)
Median	4.2 mm	4.0 mm
Macula to tumor distance		
0 mm	13.2	6 (5.2)
0.1–2.0 mm	24.9	30 (25.9)
2.1–5.0 mm	29.4	29 (25.0)
5.1–8.0 mm	16.4	26 (22.4)
> 8.0 mm	15.2	25 (21.5)
Median distance	3.0 mm	4.5 mm
Optic disc to tumor distance		
0–2.0 mm	14.9	14 (12.1)
2.1–4.0 mm	31.1	36 (31.0)
4.1–6.0 mm	24.2	27 (23.3)
6.1–8.0 mm	14.8	21 (18.1)
> 8.0 mm	14.6	18 (15.5)
Median distance	4.5 mm	4.5 mm

COMS = Collaborative Ocular Melanoma Study.

Disease outcome: local control

The definition of local failure in this study was adopted from COMS study. The COMS study (15) defined local failure as tumor expansion or extrascleral extension greater than 2 mm after brachytherapy. Protocol criteria for tumor growth were based on a two-stage observation of estimated tumor dimensions: (1) an initial increase in height of at least 15% as measured by echography or a 250-μm

Table 3
Summary of point dose and dose rate to tumor and critical structures to compare this study with COMS study

	COMS	Present study	
Points of interest	Median dose (Gy)	Median dose (Gy)	Median dose rate (cGy/h)
Apex of tumor		85.1	88.6
5-mm point		76.7	73.5
Sclera at plaque center		237.0	236.7
Center of globe		26.3	26.0
Sclera opposite plaque	7.9	7.0	6.9
Macula	79.0	50.4	50.6
Optic disc	52.1	44.7	42.3
Lens	15.6	14.5	13.7

COMS = Collaborative Ocular Melanoma Study.

Note that the data were obtained from two groups using different prescription points. The apex of tumor was used for prescription point in 108 cases, whereas the point at 5 mm above the sclera surface was used in 8 cases whose tumor height was <5 mm.

Table 4
Summary of point dose and dose rate to tumor and critical structures for 67 cases in which tumor height was <5 mm and tumor apex was used for prescription point

Points of interest	Average dose (Gy)	Average dose rate (cGy/h)
Apex of tumor	85.4	85.1
5-mm point	62.4	62.1
Sclera at plaque center	198.6	198.5
Center of globe	22.1	21.9
Sclera opposite plaque	6.4	6.4
Macula	72.1	71.2
Optic disc	48.5	48.2
Lens	12.0	11.9

The average tumor height and average implantation time was 3.5 mm and 102.3 h, respectively. The median dose to 5-mm point was 60.7 Gy.

expansion of any tumor boundary as judged from photographs or clinical examination and (2) expansion of an additional 15% increase in elevation or a further 250-μm expansion of any tumor boundary observed and confirmed on subsequent examination. In this study, the same two-stage assessment was used to ensure consecutive 15% growth in height at both stages for local failure, but the criterion of 250-μm expansion was not used. The measurement in height always came from the ultrasound and compared with that of previous photographs. The median followup for all 116 patients was 2.63 years, and 29 patients were followed for more than 5 years. The estimated local control rates (95% confidence interval [CI]) at 3, 5, and 10 years were 97.2% (94.1%–100.3%), 95.3% (90.4%–100.1%), and 95.3% (90.4%–100.1%), respectively, in the Kaplan–Meier curve (Fig. 2). Although these data look slightly more favorable when compared with the local control rate of 91% at 5 years reported by COMS (8), the 95% CI of COMS study was 89%–93% and somewhat overlapped with data in this study. Because of the small patient cohort in this study, attention should be paid in interpreting the data.

To correlate local control with multiple variables, multivariate analysis was performed for maximal tumor basal dimension and height, the presence of subretinal fluid, visual symptoms at diagnosis, the presence of orange pigment, and location of tumor <3 mm from the optic disc.

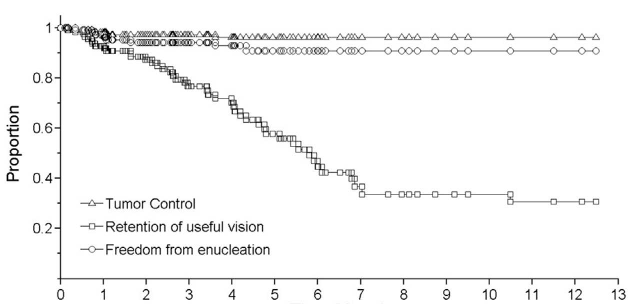

Fig. 2. Kaplan–Meier analysis for tumor control, freedom from enucleation, and retention of useful vision.

There was no statistically significant correlation between local disease control and any of these variables. In addition, multivariate analysis was performed to investigate the effect of dose and dose rate on local control for the doses at the apex of the tumor, 5 mm above the plaque, and the scleral surface above the plaque. No statistically significant correlation was found between local disease control and dose and dose rate at these dose points.

Functional outcome: eye preservation

Estimated preservation of the globe at 3, 5, and 10 years was 94.3% (89.8%–98.7%), 90.2% (83.2%–97.2%), and 90.2% (83.2%–97.2%), respectively, in the Kaplan–Meier curve (Fig. 2). The estimated preservation of the globe at 5 years in COMS study (15) was 87.5% (84.4%–90.0%). Although this study's 5 years' outcome looked slightly better than COMS study (90.2% vs. 87.5%), the 95% CI in this study was wide enough to include that of COMS data because of the small sample size. Hence, we conclude that estimated preservation of the globe at 5 years was comparable between two studies.

At the time of clinical progression, several other treatment options could be available other than enucleation, such as transpupillary thermotherapy or retreatment with external beam radiotherapy. But photodynamic therapy was not an option. None of the patients in this study received those treatment options. Instead, 8 patients eventually underwent enucleation in all. Four patients had enucleation because of tumor growth. However, there was no evidence of local failure at the time of enucleation for the other 4 patients. Reasons for enucleation despite the absence of tumor progression included pain in a blind eye (3 patients) and phthisic eye preventing tumor visualization (1 patient). In the COMS study (15), the total number of enucleation was 69 cases within 5 years. The reasons for enucleation were primarily local treatment failure (59%) followed by pain (28%), low vision acuity (9%), pain and low vision acuity (3%), suspected tumor growth (3%), and retinal detachment (1%).

According to the multivariate analysis, it was noted that tumor height was the only variable identified with a statistically significant correlation to enucleation (p = 0.031). Also, although a univariate analysis revealed that there was a correlation between enucleation and the maximum scleral dose and the dose/dose rate to eye center, a multivariate analysis was performed for the dose and dose rate to the critical structures in Table 3, and no statistically significant correlation was identified.

Outcome: vision retention

Estimated preservation of useful vision at 3, 5, and 10 years was reported: 77.7% (69.1%–86.3%), 59.3% (47.7%–70.9%), and 35.9% (22.4%–49.5%), respectively, in the Kaplan–Meier curve (Fig. 2). These data compared favorably to the COMS study, which reported 57% (52.0% −62.0%) with visual acuity better than 20/200 at 3 years (9). The 5 years' visual acuity data for better than 20/200 also showed improvement compared with COMS report no. 18 (8), 37% (32.0%−41.0%). COMS report no. 22 (16) provided 5 years' and 10 years' visual acuity data of fellow eyes using different parameters, such as improved by ≥2 lines, <2 line change, worsened by two or three lines, and worsened by >3 lines compared with the baseline data. Incidence of cataract in fellow eyes at 5 and 10 years was also reported (16).

Table 5 summarizes preservation of vision based on pretreatment baseline visual acuity. Compared with the patients with poor visual acuity (<20/40), patients with better visual acuity (>20/40) showed higher retention of vision by an average of 28.1% consistently at 3-, 5-, and 10-year followup.

Multivariate analysis was performed to investigate the correlation between vision retention and multiple disease-related variables; only tumor height demonstrated statistically significant ($p = 0.008$) correlation with vision retention. On the other hand, univariate analysis identified the correlation of vision preservation with tumor height, the presence of subretinal fluid, and the presence of visual symptoms. In addition, univariate analysis was performed to investigate the effect of dose and dose rate to the macula and optic disc. It identified a correlation between preservation of vision and the dose and dose rate to the macula. On multivariate analysis, macula dose rate remained statistically significant ($p = 0.018$).

Discussion

The present study represents a similar patient population with those reported by the COMS study. Patients reported here were slightly older with worse visual acuity at baseline, but tumor size and position and ratio of ciliary body involvement were comparable. Our outcomes with regard to local control, enucleation, and preservation of vision compared favorably with those reported by the COMS study.

This study has a limitation of short followup (median, 2.63 years). To increase median followup, 23 patients whose followup was <1 year were disregarded, and rates of local tumor control, freedom from enucleation, and retention of useful vision were re-estimated. The median followup for 93 patients was increased to 3.41 years, and the percentage of patients whose followup was >5 years was increased from 25% to 31%. The estimated local control rates at 3, 5, and 10 years were 98.9% (96.8% −101.0%), 96.9% (92.6%−101.3%), and 96.9% (92.6% −101.3%), respectively. Estimated preservation of the globe at 3, 5, and 10 years was 96.8% (93.2%−100.4%), 92.6% (85.9%−99.2%), and 92.6% (85.9%−99.2%), respectively. Estimated preservation of useful vision at 3, 5, and 10 years was 83.6% (75.4%−91.8%), 63.8% (51.7%−75.8%), and 38.7% (24.2%−53.2%), respectively. All estimated rates for those three parameters were consistently increased because the disregarded data (followup of <1 year) had low rates for those parameters.

Another limitation in this study is small sample size (116 patients). The total number of local failures was four and the latest failure at 3.61 years for local control, and eight and at 4.33 years for preservation of globe, respectively. There was no failure after 4 years for local control and after 5 years for preservation of globe, respectively. Therefore, the estimated rates at 10 years for local tumor control and preservation of globe were the same values at 5 years, and they were not meaningfully different. In contrast, there were 42 failures for vision retention within 10 years, and the latest failure occurred at 10.48 years. Hence, the estimated value at 10 years for vision retention is less disputable.

In our study, the significant deviation from the COMS treatment protocol relates to patients with tumors less than 5 mm in height. Per COMS protocol, these patients would have been prescribed 85 Gy to the 5-mm point (default point). Instead, most of these tumors in this study were prescribed to the tumor's apex point. This variation reduced the median dose values of the macula, optic disc, and lens (116 patients' data in this study), by an average of 36.2%, 14.2%, and 7.1%, respectively, compared with the COMS data (Table 3). The magnitude of dose reduction to the critical structure is dependent on their distance to the plaque. For instance, the macula dose was decreased by 28.6 Gy on average because it was the closest located critical structure to the tumor. In contrast, for the sclera opposing to the plaque (a distally located critical structure to the tumor), the dose reduction was therefore smaller (0.9 Gy compared with the data of COMS). An alternative explanation to the lower risk of vision loss (higher vision retention) noted in this study is that tumors in this study were further from the macula (31.1% within 2 mm of macula), than what was observed in the COMS (38.1% within 2 mm of macula) (Table 2). As such, the prescription point difference may have nothing to do with the fact that a group of patients with tumors closer to the macula are likely to have worse visual outcomes.

Although retrospective, this study suggested that small tumors could be safely prescribed to the apex of the tumor without reduction in local control. The change in prescription point from 5-mm point to tumor apex for tumors < 5 mm thick allows for the reduction of dose to sensitive

Table 5
Percent of patients for vision retention at 3, 5, and 10 years of followup

Baseline visual acuity	Retention of vision (%) 3 y	5 y	10 y
≥20/40	88.7	64.7	49.4
<20/40	59.2	45.5	13.7

structures and could result in improved visual outcomes. In a study reported by Saconn *et al.* (17), the prescription dose was reduced down to 63 Gy (median) to achieve better visual outcomes. Importantly, 73% of the tumors in that study were 5 mm or less in tumor height and were prescribed to the 5-mm point. The authors reported 84% preservation of useful vision at 3 years and a local control rate of 91% at 5 years. In our study, the median dose at 5 mm was 76.7 Gy, but for 75 patients with tumors 5 mm or less in height, the median and average dose values were 61.4 and 64.8 Gy comparable to those (63.0 and 62.5 Gy) reported by Saconn *et al*. Furthermore, if 8 patients who were prescribed to the 5 mm above the sclera were excluded, the median and average dose values for remaining 67 patients were 60.7 and 62.4 Gy, respectively (Table 4). Prescribing to the tumor apex for smaller lesions has a similar dosimetric impact to dose reduction. Recently, very similar data to ours were published from a study (18) in which the same approach as prescribing to the apex of tumor was used. Local tumor control at 5 years was 91% comparable to COMS data. They concluded that increasing apex and 5-mm dose were correlated with greater visual acuity loss and radiation complication.

In the COMS study manual (19), dose calculation was recommended to use the original AAPM 43 formalism (11) with point source approximation while anisotropic factor was ignored. In this study, the radial dose function was defined as close as 1 mm to the source by adapting updated TG-43 parameters. This is more accurate than the original TG-43 formalism whose radial dose function is defined up to 5 mm from the source, and it is not appropriate for eye plaque dosimetry because of steep dose gradient (i.e., within millimeters from the source). However, the accuracy of dose calculation in this study is still limited because of point source approximation and ignorance of heterogeneity. TG-43 formalism assumed homogenous water medium with full scatter condition by disregarding any perturbation because of heterogeneity of eye structure (i.e., bony orbit) and plaque material such as gold–alloy backing and silastic insert. Even radioactive seeds themselves can change an intended dose distribution, which is called interseed effect. Therefore, the AAPM TG-129 (20) investigated all these effects on eye plaque dosimetry and pointed out that prescribing 85 Gy to 5 mm with ^{125}I sources in the TG-43 formalism delivered a real dose of 75 Gy if all heterogeneity is taken into account. For clinical eye plaque dosimetry, it was recommended that line source approximation in homogeneous water phantom and two-dimensional TG-43U1 dosimetry should be used until a suitable commercial Monte Carlo–based brachytherapy TPS is available. Hence, it is noted that dose data reported in this study should not be taken as absolute but be interpreted in comparison with COMS data. The absolute dose can be only accurately computed with advanced dose calculation algorithm such as Monte Carlo after taking all heterogeneity effects discussed in TG-129 into account.

For 100 patients, dose calculation was repeated with line source approximation of TG-43U1 to follow TG-129 recommendation, and dose to each structure was compared with that with point source approximation (21). Average (maximum) relative dose difference from point source approximation was 0.7% (1.6%) at apex of tumor, 0.7% (1.3%) at 5-mm point, 1.2% (2.1%) for sclera at plaque center, 0.5% (0.9%) at center of globe, 0.3% (0.6%) at sclera opposite plaque, 2.1% (8.1%) at macula, 1.2% (2.7%) at lens, and 1.6% (7.0%) at optic disc, respectively.

In the clinic, the duration of implant is not exactly the same as planned because of scheduling variation in the operating room at the implant date. For most patients, the implantation and removal times varied within ±1 h with maximum of <±3 h relative to the scheduled time. Hence, the worst scenario could be 6-h difference (3.6% of 168-h implantation) from the planned, that is 3-h delayed implantation and 3-h early removal. In addition, the radioactive source strength delivered from the vendor was slightly different from that ordered, depending on the available source activity at the time of ordering. The difference of activity between ordered and delivered seeds was <±3% in most cases and always <±5%. Therefore, in theory, the worst dose discrepancy could be 8.6% if 6-h implantation time difference and 5% source strength difference occurred simultaneously. Recently, postplan dosimetry has been implemented in the clinic, and recomputed doses based on the delivered source activity and actual implantation time were reported. For 33 recent patients, mean ± standard deviation (maximum) value of dose discrepancy from the prescribed dose of 85 Gy was 0.7% ± 0.9% (3.3%).

The dose rate in the present study was different from the American Brachytherapy Society (ABS) recommendations (22, 23). Current ABS guidelines recommend the dose rate at the prescription point to range between 60 and 105 cGy/h. COMS allowed prescription dose rate as low as 42 cGy/h. This is of logistical importance as many surgeons have weekly allocations of operating room time. An implant of 7 days corresponds to a prescription point dose rate of 50.6 cGy/h. The ABS recommendation is based on a retrospective review reported by Quivey *et al.* (10). The authors reported a significant correlation between local control and dose rate at the tumor apex. However, their prescription range was wide from 30 to 141 Gy. In contrast, the present study was more homogeneous with regard to the prescription dose. The vast majority of patients (108/116) were prescribed 85 Gy to the tumor apex, and the remaining 8 patients were prescribed 85 Gy to 5 mm above the plaque. Although prescribed with a consistent total dose, patients in the present study were treated with variable dose rate. This allows for a more rigorous analysis of the effect of dose rate on tumor control. In this study, no correlation was found between variation of apex dose rate (ranging from 50.6 to 98.2 cGy/h) and tumor control. This suggests that a modest reduction in dose rate to 50.6 cGy/h is clinically feasible without compromising local control.

Conclusion

The clinical outcomes data of this retrospective single-institution study confirm what has been known for 3 decades from the COMS study that ^{125}I episcleral plaque therapy is an effective, low morbidity, treatment modality for medium-sized choroidal melanomas. For tumors with a height less than 5 mm, reducing the prescription depth to the tumor apex (instead of 5 mm) enabled us to decrease the dose to all sensitive structures within the eye. This dose reduction was feasible without any loss in local control. Although the dose rate varied from the ABS guidelines because of limited availability of operating room (i.e., weekly), there was no difference in either local tumor control probability or complication rates.

Acknowledgments

The authors thank Mr. Vi Nhan Nguyen, MS, for providing dose comparison data between point and line source approximations of TG-43U1.

References

[1] Kim RY, Gragroudas ES, Young LHY. Future laser approaches in melanoma treatment. *Int Ophthalmol Clin* 2006;46:27–39.

[2] Daftari IK, Petti PL, Shrieve DC, *et al*. Newer radiation modalities for choroidal tumors. *Int Ophthalmol Clin* 2006;46:69–79.

[3] Char DH, Phillips T, Daftari I. Proton teletherapy of uveal melanoma. *Int Ophthalmol Clin* 2006;46:41–49.

[4] Hui JI, Murray TG. Radioactive plaque therapy. *Int Ophthalmol Clin* 2006;46:51–68.

[5] Damato B. The role of eyewall resection in uveal melanoma management. *Int Ophthalmol Clin* 2006;46:81–93.

[6] Mueller AJ, Talies S, Schaller UC, *et al*. Stereotactic radiosurgery of large uveal melanomas with the gamma-knife. *Ophthalmology* 2000;107:1381–1387. discussion 1387–1388.

[7] Factors predictive of growth and treatment of small choroidal melanoma: COMS Report No. 5. The Collaborative Ocular Melanoma Study Group. *Arch Ophthalmol* 1997;115:1537–1544.

[8] Diener-West M, Earle JD, Fine SL, *et al*. The COMS randomized trial of iodine 125 brachytherapy for choroidal melanoma, III: initial mortality findings. COMS Report No. 18. *Arch Ophthalmol* 2001;119:969–982.

[9] Melia BM, Abramson DH, Albert DM, *et al*. Collaborative ocular melanoma study (COMS) randomized trial of I-125 brachytherapy for medium choroidal melanoma. I. Visual acuity after 3 years COMS report no. 16. *Ophthalmology* 2001;108:348–366.

[10] Quivey JM, Augsburger J, Snelling L, Brady LW. 125I plaque therapy for uveal melanoma: Analysis of the impact of time and dose factors on local control. *Cancer* 1996;77:2356–2362.

[11] Nath R, Anderson LL, Luxton G, *et al*. Dosimetry of interstitial brachytherapy sources: Recommendations of the AAPM Radiation Therapy Committee Task Group No. 43. American Association of Physicists in Medicine. *Med Phys* 1995;22:209–234.

[12] Rivard MJ, Coursey BM, DeWerd LA, *et al*. Update of AAPM Task Group No. 43 Report: A revised AAPM protocol for brachytherapy dose calculations. *Med Phys* 2004;31:633–674.

[13] Rivard MJ, Butler WM, DeWerd LA, *et al*. Supplement to the 2004 update of the AAPM Task Group No. 43 Report. *Med Phys* 2007;34:2187–2205.

[14] Coordinate localization of eye plaque for Collaborative Ocular Melanoma Study (COMS). Available at: http://rpc.mdanderson.org/rpc/credentialing/COMS.htm. Accessed October 23, 2014.

[15] Jampol LM, Moy CS, Murray TG, *et al*. The COMS randomized trial of iodine 125 brachytherapy for choroidal melanoma: IV. Local treatment failure and enucleation in the first 5 years after brachytherapy. COMS Report No. 19. *Ophthalmology* 2002;109:2197–2206.

[16] Ten-year follow-up of fellow eyes of patients enrolled in Collaborative Ocular Melanoma Study randomized trials: COMS report no. 22. *Ophthalmology* 2004;111:966–976.

[17] Saconn PA, Gee CJ, Greven CM, *et al*. Alternative dose for choroidal melanoma treated with an iodine-125 radioactive plaque: A single-institution retrospective study. *Int J Radiat Oncol Biol Phys* 2010;78:844–848.

[18] Perez BA, Mettu P, Vajzovic L, *et al*. Uveal melanoma treated with iodine-125 episcleral plaque: An analysis of dose on disease control and visual outcomes. *Int J Radiat Oncol Biol Phys* 2014;89:127–136.

[19] COMS manual – Chapter 12. Radiation therapy. Available at: http://pages.jh.edu/wctb/coms/manual/coms_chap12.pdf. Accessed January 23, 2015.

[20] Chiu-Tsao ST, Astrahan MA, Finger PT, *et al*. Dosimetry of (125)I and (103)Pd COMS eye plaques for intraocular tumors: Report of Task Group 129 by the AAPM and ABS. *Med Phys* 2012;39:6161–6184.

[21] Nguyen VN, Cheung MK, Vonk D, *et al*. Evaluate and compare the doses differences between line source and point source calculation methods for episcleral plaque brachytherapy. [Abstract]. *Brachytherapy* 2014;13:S101.

[22] Nag S, Quivey JM, Earle JD, *et al*. The American Brachytherapy Society recommendations for brachytherapy of uveal melanomas. *Int J Radiat Oncol Biol Phys* 2003;56:544–555.

[23] ABS-OOTF Committee. The American Brachytherapy Society consensus guidelines for plaque brachytherapy of uveal melanoma and retinoblastoma. *Brachytherapy* 2014;13:1–14.